

Generation Interconnection System Impact Study Report

For Queue Project Q18-1108

Hockinson 100 kV

March 04, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1108. The Interconnection Customer proposes a 150.0 MW generating facility interconnecting at the Hockinson 100 kV substation in the Illinois area. The request was received on November 21, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 403 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,210,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Hockinson / San Leandro U 100 kV L: 118.6% loading (102 MVA) following the outage of the Hockinson / Bsalt 100 kV L; DFAX = 58.5%.

Jacobs Corner / Pearsall 100 kV L: 112.8% loading (111 MVA) following the outage of the Kinsington / River Rock 100 kV; DFAX = 20.8%.

For the loss of Carbon Center 500 kV/345 kV Autotransformer, the Blackrock / Caldwell 345 kV loads to 103.6% of its 1,624 MVA rating. Project DFAX: 5.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hockinson Substation / San Leandro U 100 kV	Rebuild line with higher-capacity conductor	\$10,043,000	77% (\$7,753,000)
Jacobs Corner / Pearsall 100 kV	Rebuild line with higher-capacity conductor	\$5,541,000	22% (\$1,239,000)
Caldwell / Blackrock 345 kV	Reconductor line	\$20,520,000	1% (\$218,000)
	Total allocated to project		\$9,210,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.